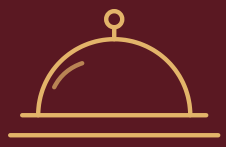
ROLE BRIEF · RESTAURANT BILLING SAAS



Head Chef

Gagan

Full Stack + DevOps Engineer · Tech Lead

A backend and offline engine that never loses an order or double-charges a bill, deployed and monitored in production.

SERVICE WINDOW	PLATES	FINAL TASTING
13 Oct – 1 Nov 2026	Standard · Pro · Premium	AC-01 to AC-12

Works with **Aditya** · Sous Chef (Frontend & UI)

Companion to **The Kitchen Playbook**

BASED ON SRS V1.0 · PREPARED 12 OCTOBER 2026

COURSE ONE

01 Your Role: Head Chef

You own everything behind the pass: data, cloud API, branch hub, sync, security and infrastructure, plus the client relationship. Your daily job is to keep Aditya one contract ahead and keep the money paths provably correct.

REPORTS TO	WORKS WITH	TIME SPLIT
Client (you are the lead)	Aditya, Sous Chef	70 build / 15 review / 15 ops + client

Mission. A backend and offline engine that never loses an order or double-charges a bill, deployed and monitored in production.

What good looks like on 1 Nov

0 lost or doubled orders
in the offline disconnect/reconnect harness

12 of 12 AC tests green
in CI before the final demo

Contract by 12:00 daily
Aditya never waits on an endpoint

PR review under 2 hours
during working hours

Every merge on staging
auto-deploy from `main`

Restore drill logged
before release candidate on Day 18

Your decision rights: Architecture, data model, security, release go/no-go, and scope cuts after the client agrees.

Daily capacity



70% build · 15% review / unblocking Aditya · 15% DevOps + client

COURSE TWO

02 What You Own

> Everything below is yours to build and answer for. Items marked as review are checked by Aditya before they merge.

Architecture & contracts

- Architecture doc, DB schema, permission matrix (role × action × plan)
- OpenAPI contract + mock JSON for every module, published by 12:00 the day before Aditya needs it
- Shared Dart `core` package: models, money in paise, GST, rounding, golden test vectors

Cloud API (NestJS + Postgres)

- Tenants, restaurant lifecycle, Postgres row-level security on `tenant_id`
- Auth (Argon2, JWT 15 min + rotating refresh), RBAC guards, platform admins
- MFA: TOTP, hashed recovery codes, step-up for reset, notification + audit
- Plans, licenses, `@RequiresFeature()` guards, Ed25519-signed license token
- Append-only audit service; franchise/branches; auditor read-only; reports; franchise invoice data

Branch hub & sync (Dart)

- `shelf` server inside the Windows app, SQLite via `drift`, WebSocket event bus
- QR pairing with pinned TLS, mDNS discovery, device tokens
- Orders, bills, payments with idempotency keys, row-version concurrency, gap-free invoice numbers
- Outbox push, cloud dedupe, config pull by cursor, `SyncConflict`, reconciliation report
- ESC/POS printing over TCP 9100 + Windows PDF fallback
- KDS events, recipe versions, approvals, inventory movements, shifts

DevOps

- Docker Compose + Caddy on India-region cloud; managed Postgres; S3/R2 storage
- GitHub Actions: lint, tests, API image, Windows installer, Android APK/AAB, Flutter Web deploy
- Sentry, Uptime Kuma, JSON logs; daily backups + PITR; restore drill logged
- Code-signing certificate ordered Day 1; secrets in CI, never in the repo

Quality & client

- Automated tests for entitlement, RBAC, tenant isolation, money math, offline reconnect, duplicate payment
- Review every backend PR, including Aditya's CRUD; merge authority on `main`
- Daily 3-line client note; checkpoint demos; written sign-offs; final invoice

COURSE THREE

03 Your Stations (RACI)

> 22 of 24 modules need work from you. A = you answer for it, R = you do the work; Aditya's part is shown so you know who to talk to.

Module	You	Aditya	Done by
Architecture, schema, API contract, permission matrix	**A/R**	C	Day 2
Tenant onboarding, restaurant lifecycle	**A/R backend**	R portal UI	Day 3
Auth, RBAC, platform admins, MFA + step-up reset	**A/R**	R screens	Day 4
Licensing, plans, entitlements	**A/R**	R screens	Day 4
Audit & security logs	**A/R**	R viewer	Day 4

Module	You	Aditya	Done by
Menu, categories, tax rules, images	**A, review**	R (CRUD + UI)	Day 6
Tables, floors, table transfer	**A/R transfer logic**	R CRUD + layout UI	Day 9 / 11
Orders, bills, taxes, tips, discounts, reversals	**A/R**	R screens	Day 9
Printing	**A/R driver**	R receipt layout	Day 9
Multi-cashier concurrency, duplicate-payment guard	**A/R**	C	Day 11
Local LAN hub, pairing, offline queue, cloud sync	**A/R**	R status UX	Day 13
Franchise/branches, branch-scoped RBAC, auditor	**A/R**	R dashboards	Day 14
Kitchen Display System	**R events**	A/R screens	Day 15
Recipes + menu approval workflow	**A/R**	R screens	Day 15
Inventory + stock lifecycle	**A/R**	R screens	Day 16
Shifts, breaks, attendance	**R**	A/R	Day 17
Consolidated franchise invoice	**A/R**	R PDF layout	Day 17
Reports & dashboards	**R queries**	A/R	Day 17
Security review, load + offline tests	**A/R**	C	Day 18
CI/CD, infra, backups, monitoring, installers	**A/R**	I	Day 19
User docs + admin ops guide	**A/R ops guide**	A/R user docs	Day 19
Client demos, sign-offs, invoice	**A/R**	C	Each checkpoint

04 COURSE FOUR
Your 20-Day Service

> One row per day, 13 Oct to 1 Nov 2026. Tick each row at the 20:00 Closing Check. Shaded rows are client tastings.

Day	Date	Your tasks	Contract drop by 12:00 (for Aditya's tomorrow)	Tasting / gate	Done
1	Tue 13 Oct	Kickoff call; send Chef's Defaults for sign-off; monorepo, CI skeleton, staging server, Postgres; schema v1	Login / auth API contract + mocks	Defaults sent to client	☐
2	Wed 14 Oct	Architecture doc, permission matrix, OpenAPI v1; auth (Argon2, JWT, refresh), tenants, users, RBAC guards, seed	Tenant + onboarding contract	Handoff pack to client; defaults signed	☐
3	Thu 15 Oct	MFA (TOTP, recovery codes, step-up), platform admins, audit service (append-only)	License, admin-roles, MFA, audit contracts		☐
4	Fri 16 Oct	Plans, licenses, entitlements guard, signed license token, suspend/block/grace	Hub pairing API + WebSocket event spec		☐
5	Sat 17 Oct	Hub server skeleton (shelf + drift), QR pairing, WebSocket bus, license verify on hub	Menu CRUD module pattern for Aditya to copy	**Course 1: AC-01, 02, 03 demo**	☐
6	Sun 18 Oct	Config pull sync (menu, tax, users, tables) cloud to hub; review Aditya's CRUD	Order engine contract + table status events		☐
7	Mon 19 Oct	core billing math + golden vectors; order engine on hub (lines, price snapshots, status history)	Bill / payment / tip / reversal contracts		☐

Day	Date	Your tasks	Contract drop by 12:00 (for Aditya's tomorrow)	Tasting / gate	Done
8	Tue 20 Oct	Bills, payments with idempotency, discounts (permission), tips to waiter, reversals with reason, bill adjustments	Table transfer + print API		☐
9	Wed 21 Oct	Table transfer (atomic, conflict check); ESC/POS + PDF printing	Sync status API		☐
10	Thu 22 Oct	Outbox push sync, cloud dedupe, reconciliation report, sync status API	Floor model + concurrency events	**Course 2: Standard MVP, AC-04 to 06; build sent**	☐
11	Fri 23 Oct	Floors, multi-cashier concurrency (row versions), duplicate-payment tests	Waiter-scoped order APIs		☐
12	Sat 24 Oct	Resilience: hub restart recovery, client retry queue, Wi-Fi drop/reconnect	Connection-state events		☐
13	Sun 25 Oct	Offline test harness: WAN cut, LAN cut, duplicate + conflict simulation	KDS ticket event spec	**Course 3: Professional, AC-07, 08**	☐
14	Mon 26 Oct	Franchise + branches, branch-scoped RBAC, auditor read-only (UI + API)	Recipe + approval API		☐
15	Tue 27 Oct	KDS events + timestamps; recipe versions; menu approval workflow	Inventory API		☐
16	Wed 28 Oct	Inventory: departments, items, receipts, transfers, adjustments, wastage, recipe deduction at Ready, low-stock alerts	Shift API, report queries, invoice data		☐
17	Thu 29 Oct	Shifts/breaks API, consolidated franchise invoice (PDF), report queries	Security-pass bug list	**Course 4: Premium, AC-09 to 12**	☐
18	Fri 30 Oct	Security pass (RLS tests, rate limits, secret scan, dependency audit), k6 load test, backup + restore drill	Production URLs + build pipeline	Release candidate rc1	☐
19	Sat 31 Oct	Production infra, signed Windows installer, Android AAB/APK, monitoring + alerts, ops guide	UAT fix list, split by layer	**Client UAT on production**	☐
20	Sun 1 Nov	UAT fixes, final full AC-01 to AC-12 demo, handoff pack, invoice		**Sign-off + final invoice**	☐

Buffer: Mon 2 Nov to Sat 7 Nov for slips and client-requested fixes. Cut line: if Course 3 is not green by Sun 25 Oct, Premium drops to the reduced scope in the main playbook.

COURSE FIVE

05 Your Quality Checks at the Pass

> You own 12 of the 12 acceptance criteria the client signs before paying the balance. Each needs a passing test and a live demo on its day.

AC	Workflow	Demo day	Automated test you write (happy + failure path)	Shared with
AC-01	Tenant onboarding	5	Verified email to tenant + license; unverified or duplicate email rejected; tenant A sees 0 rows of tenant B	
AC-02	License enforcement	5	Expired/blocked tenant gets 403; tampered license token rejected by hub	
AC-03	MFA reset	5	Unauthorised admin 403; reset without step-up rejected; success writes audit + notification	
AC-04	Table transfer	10	Order moves once with all lines; occupied target conflicts; two simultaneous transfers, one wins	
AC-05	Item reversal	10	No reason rejected; accepted reversal creates adjustment + audit, updates totals	
AC-06	Tip attribution	10	Tip saved to selected waiter; appears in tip report	Aditya (UI)
AC-07	Concurrent cashiers	13	Two terminals consistent within 1 s; same idempotency key twice, one payment	
AC-08	Offline LAN	13	WAN cut: waiter order reaches cashier via hub; reconnect pushes once	
AC-09	KDS	17	Ticket appears; accept, preparing, ready visible to waiter and cashier with timestamps	Aditya (UI)
AC-10	Recipe inventory	17	Ready ticket deducts mapped ingredients; reversal after Ready creates wastage	
AC-11	Auditor	17	Auditor sees authorised records; every write endpoint 403 for auditor	
AC-12	Franchise invoice	17	Monthly invoice shows per-branch lines and consolidated total	Aditya (UI)

COURSE SIX

06 Keeping Aditya unblocked

1. Publish tomorrow's contract + mock JSON in `packages/contracts` by 12:00 (the Contract drop column below).
2. Give Aditya one finished module (menu CRUD) on Day 5 as the pattern for his light backend work, then review his CRUD on Day 6.
3. Treat contract changes as PRs tagged for both of you, merged the same day.
4. When a bug's layer is unclear, triage it within 2 hours and assign it.
5. Never let Aditya work around the API: if he raises a change request, answer it the same day.

Daily rhythm

Time (IST)	Ritual	What you do
10:00	Morning Briefing (15 min)	Yesterday done, today's plate, blockers
12:00	Contract drop	Publish tomorrow's contract + mocks for Aditya
18:30	Staging tasting	Click through today's features with Aditya

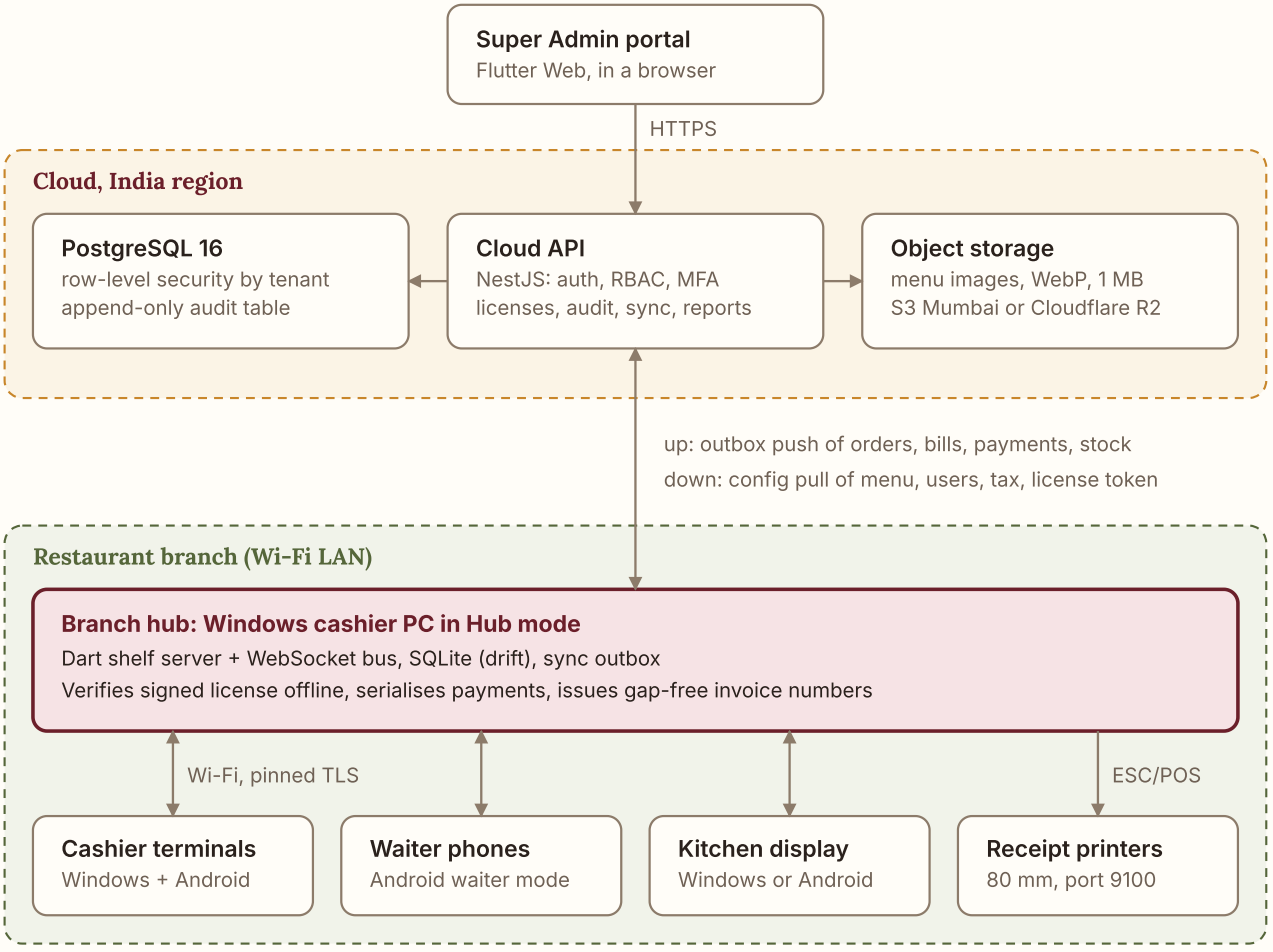
Time (IST)	Ritual	What you do
20:00	Closing Check	Push, update the board, send the client a 3-line note

Definition of done: merged to `main`; on staging; happy-path and one failure-path test pass; critical actions have audit + permission checks; screens have loading, empty and error states; board card moved to Done.

COURSE SEVEN

07 Your Kitchen and Tools

You design and run all of this. The hub is the single writer for live operations in each branch; the cloud is the single writer for configuration.



Branch devices talk to the hub; only the hub talks to the cloud. If the hub is switched off, the branch pauses rather than splitting into two versions of the truth.

Your stack

Layer	Choice	Why it fits 20 days	Owner
Branch hub server	Dart `shelf` + WebSocket, embedded in the Windows app ("Hub mode")	Shares the billing core with clients; no separate install	Gagan

Layer	Choice	Why it fits 20 days	Owner
Local storage	SQLite via `drift` (hub and clients)	Reliable local-first persistence, migrations, typed queries	Gagan
Shared core	Dart package `core`: models, money (integer paise), GST, validation	Same totals on every device; golden test vectors	Gagan
Cloud API	NestJS (TypeScript) + Prisma	Fast CRUD, guards for RBAC/entitlements, OpenAPI generation	Gagan (Aditya: simple CRUD)
Database	PostgreSQL 16, row-level security on `tenant_id`	Tenant isolation enforced in the DB, not only in code	Gagan
Realtime	WebSocket on hub (LAN); Socket.IO on cloud for portal	Live order/table/KDS state across terminals	Gagan
Auth / MFA	Argon2 hashing, JWT (15 min) + rotating refresh, TOTP (`otplib`) + hashed recovery codes	Covers SRS §3.3 and §9 directly	Gagan
License token	Ed25519-signed token (plan, features, limits, expiry), verified offline on hub	Satisfies "do not trust a local license flag"	Gagan
Files / images	S3-compatible storage (AWS S3 Mumbai or Cloudflare R2), 1 MB cap, WebP	Cheap, simple signed uploads	Gagan
Printing	ESC/POS over TCP 9100 (80 mm thermal) + Windows PDF / system print fallback	Works on both OSes without vendor SDKs	Gagan + Aditya
Hosting	AWS ap-south-1 (EC2/Lightsail + RDS) or DigitalOcean Bangalore; Docker Compose + Caddy (auto TLS)	India data residency, one-person ops	Gagan
CI/CD	GitHub Actions: lint, tests, API image, Windows installer (MSIX / Inno Setup), Android APK/AAB, Flutter Web deploy	Every merge to `main` lands on staging	Gagan
Observability	Sentry (API + Flutter), Uptime Kuma, structured JSON logs	Errors visible during UAT	Gagan
Testing	Jest + Supertest (API), Dart `test` (core/hub), Flutter widget + integration tests, k6 (load)	Covers SRS §9 automated test list	Both

COURSE EIGHT

08 Your Kitchen Fires

Risk	Likelihood	What you do about it
Client delays sign-off on defaults or scope	High	Send Day 1, deadline Day 2, state in writing that silence means approval
Offline sync bugs (duplicates, lost orders)	Medium	Single writer per branch, idempotent events, sync on from Day 10, harness from Day 13
You become the review bottleneck	High	Contracts a day ahead; 2-hour review SLA; Aditya builds on mocks
Printer quirks on Flutter Windows	Medium	Test the client's actual printer model by Day 9; PDF fallback
Scope creep during UAT	High	New asks go to the v1.1 list unless they are bugs against the playbook
Unsigned installer blocked by SmartScreen	Medium	Order a code-signing certificate on Day 1 or ship MSIX with sideload steps
Burnout	High	Take your half-day off Sun 18 Oct PM and one more in week 3

COURSE NINE

09 Your Closing Checklist

Before full build, Day 2

- [ ] Architecture diagram
- [ ] API contract / OpenAPI v1
- [ ] Database schema
- [ ] Permission matrix
- [ ] Chef's Defaults + scope table signed by client

Handoff pack, Day 20

- [ ] Repo access shared with client
- [ ] Deployment instructions + infrastructure-as-code
- [ ] Backup / restore procedure with logged drill
- [ ] Monitoring and alerts with client access
- [ ] Admin operations guide
- [ ] Windows installer, Android APK/AAB, portal URL
- [ ] Test report: all AC tests + offline simulations
- [ ] v1.1 backlog agreed

Gagan, Head Chef · Signature & date

Aditya, Sous Chef (Frontend & UI) · Acknowledged